

Exponential distribution

A continuous random variable 'X' is said to follow exponential distribution if

$$f(x) = \begin{cases} \theta e^{-\theta x}, & x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

Moment generating function:

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \int_0^{\infty} e^{tx} f(x) dx \\ &= \int_0^{\infty} e^{tx} \theta e^{-\theta x} dx \\ &= \theta \int_0^{\infty} e^{-x(\theta - t)} dx \\ &= \theta \left[\frac{e^{-x(\theta - t)}}{-(\theta - t)} \right]_0^{\infty} \\ &= \frac{\theta}{\theta - t} = \theta (\theta - t)^{-1} \end{aligned}$$

Mean and variance of exponential distribution

$$\begin{aligned} \mu_1^1 &= \frac{d}{dt} [M_X(t)]_{t=0} \\ &= \frac{d}{dt} [\theta(\theta - t)^{-1}]_{t=0} \\ &= [\theta(-1)(\theta - t)^{-2}(0 - 1)]_{t=0} \\ &= [\theta(\theta - t)^{-2}]_{t=0} \\ &= \theta\theta^{-2} \\ &= \frac{1}{\theta} \end{aligned}$$

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$$\begin{aligned}
\mu_2^1 &= \frac{d^2}{dt^2} [M_x(t)]_{t=0} \\
&= \frac{d}{dt} [\theta(\theta - t)^{-2}]_{t=0} \\
&= [\theta(-2)(\theta - t)^{-3}(0 - 1)]_{t=0} \\
&= [2\theta(\theta - t)^{-3}]_{t=0} \\
&= 2\theta.\theta^{-3} \\
&= \frac{2}{\theta^2}
\end{aligned}$$

Variance of exponential distribution:

$$\mu_2 = \mu_2^1 - (\mu_1^1)^2 = \frac{1}{\theta^2}$$

Problems:

1. The mileage which car owners get with a certain kind of radial tire is a random variable having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last

1)at least 20,000 km 2)at most 30,000km

Let X denotes the mileage obtained with the tire.

$$f(x) = \theta e^{-\theta x}$$

$$f(x) = \frac{1}{40000} e^{-\frac{x}{40000}}, x > 0 \quad \because \text{Mean} = \frac{1}{\theta} = 40,000 \Rightarrow \theta = \frac{1}{40000}$$

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1)

$$\begin{aligned} p(x \geq 20000) &= \int_{20000}^{\infty} \frac{1}{40000} e^{-\frac{x}{40000}} dx \\ &= \frac{1}{40000} \left[\frac{e^{-\frac{x}{40000}}}{-\frac{1}{40000}} \right]_{20000}^{\infty} = e^{-0.5} = 0.6065 \end{aligned}$$

2)

$$\begin{aligned} p(x \leq 30000) &= \int_0^{30000} \frac{1}{40000} e^{-\frac{x}{40000}} dx \\ &= \frac{1}{40000} \left[\frac{e^{-\frac{x}{40000}}}{-\frac{1}{40000}} \right]_0^{30000} = 1 - e^{-0.75} = 0.5270 \end{aligned}$$

Result:

Memoryless property of the exponential distribution

If X is exponentially distributed, then

$$P(X > s + t / x > s) = P(X > t), \text{ for any } s, t > 0$$

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Problem:

1. The time(in hours) required to repair a machine is exponentially distributed with parameter $\theta = \lambda = \frac{1}{2}$.

a) What is the probability that the repair time exceeds 2h?

b) What is the conditional probability that a repair takes at least 10 h given that its duration exceeds 9h?

If X represents the time to repair the machine, the density function of X is given by

$$f(x) = \theta e^{-\theta x} = \lambda e^{-\lambda x} = \frac{1}{2} e^{-\frac{x}{2}}$$

a)

$$P(X > 2) =$$

$$\int_2^{\infty} \frac{1}{2} e^{-\frac{x}{2}} dx = \left(-e^{-\frac{x}{2}} \right)_2^{\infty} = e^{-1} = 0.3679$$

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b)

$$P(x \geq 10 / x > 9) = P(x > 1)$$

$$[U \text{ sin g result : } P(x \geq 9 + 1 / x > 9) = P(x > 1)]$$

$$= \int_1^{\infty} \frac{1}{2} e^{-\frac{x}{2}} dx = \left(-e^{-\frac{x}{2}} \right)_1^{\infty} = e^{-0.5} = 0.6065$$

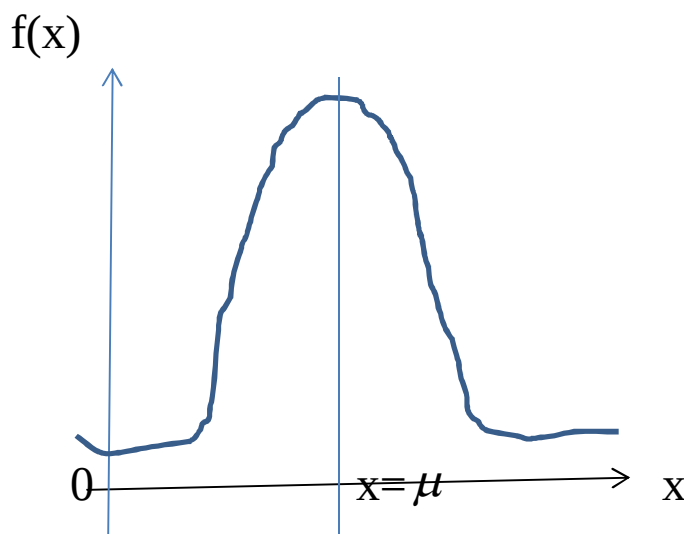
Normal Distribution: [Gaussian distribution]

A normal distribution is a continuous distribution given by

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{where } X \text{ is a continuous normal variate}$$

distributed with density function $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ with mean μ and standard deviation σ .

Normal probability curve:



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The curve is symmetrical about the ordinate $x=\mu$. The ordinate $f(x)$ decreases rapidly as x increases numerically, the maximum (occurring at $x=\mu$) given by $\frac{1}{\sigma\sqrt{2\pi}}$.

Moment generating function, Mean, Variance and Mode of the Normal distribution:

$$\text{MGF} = M_x(t) = e^{\mu t} + \frac{1}{2}t^2\sigma^2$$

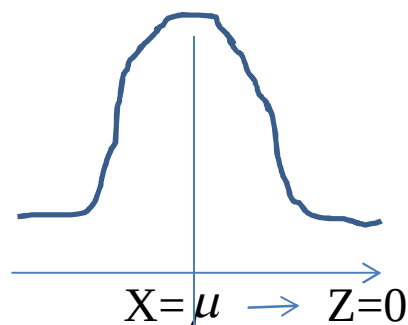
If X follows $N(\mu, \sigma)$ then $\mu_1^1 = \mu$ and variance $\mu_2 = \sigma^2$,
Mode= μ , Median= μ .

Application:

Normal distribution is applied to a large extent in statistical quality control in industry.

Standard normal probability distribution

If X is a normally distributed random variable μ and σ are respectively its mean and standard deviation, then $z = \frac{x-\mu}{\sigma}$ is called standard normal random variable.



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Note: The total area under the curve is unity. i.e., $\int f(x)dx = 1$

Problem:

1. X is normally distributed and the mean of X is 12 and standard deviation is 4. Find out the probability of the following a) $X \geq 20$ b) $X \leq 20$ c) $0 \leq X \leq 12$

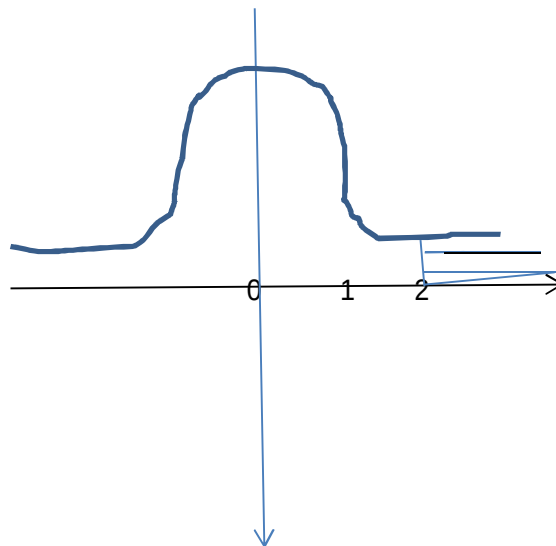
given $\mu = 12$; $\sigma = 4$

a) To find $P(X \geq 20)$

$$\text{When } X = 20, z = \frac{X - \mu}{\sigma} = \frac{20 - 12}{4} = 2$$

when $X = 20$, $z = 2$

$$\therefore P(X \geq 20) = P(Z \geq 2)$$



$$= 0.5 - P(0 \leq z \leq 2)$$

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$$= 0.5 - 0.4772$$

$$= 0.0228$$

b)To find

$$P (X \leq 20)$$

$$\text{When } X = 20 ; Z = 2$$

$$\therefore P (X \leq 20) = P (Z \leq 2)$$

$$= 1 - P (Z \geq 2)$$

$$= 1 - 0.0228$$

$$= 0.9772$$

c)To Find

$$P(0 \leq X \leq 12)$$

$$\text{When } X = 0, z = \frac{X - \mu}{\sigma} = \frac{0 - 12}{4} = -3$$

$$\text{When } X = 12, z = \frac{12 - 12}{4} = 0$$

$$\therefore P(0 \leq X \leq 12) = P(-3 \leq z \leq 0)$$

$$= P(0 \leq z \leq 3)$$

$$= 0.4987$$

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